

Boundary-Aligned Cyclic-Ulam Structure and Phase-Local Leakage

An exact finite-dimensional edge at a postcritically finite quadratic map

RH research program

August 2026

Abstract

We isolate an exact finite-dimensional mechanism in a postcritically finite quadratic map. At the algebraic parameter u_c satisfying $u_c^3 - 2u_c^2 + 2u_c - 2 = 0$, the invariant interval splits into two bands that are exchanged by one iterate. For every finite exact cell-overlap Ulam partition whose boundary contains the band endpoint $r = u_c - 1$, the matrix is anti-diagonal in band coordinates and inherits a sign vector with eigenvalue -1 . If one cell crosses the endpoint, the projected sign has the exact local defect $4\theta(1 - \theta)$, or $4h\theta(1 - \theta)$ after weighting by the cell width. A frozen 33-phase scan at four resolutions reproduces the predicted aligned/crossing dichotomy and records the finite near- -1 drift. The scan is a diagnostic: no universal power law, continuum isolated-spectrum theorem, canonical arithmetic operator, or Riemann-Hypothesis implication is claimed. An explicit overlap ledger separates this edge from RH-3, RH-10, and RH-55.

1 Scope and claim boundary

The object of this paper is a finite exact Ulam matrix, not an unnamed continuum transfer operator. We use the phrase “cyclic mode” for the two-band exchange and “leakage” for a projected finite-cell diagnostic. The following statements are outside the paper’s claim ceiling: a strong-Banach-space isolated eigenvalue or a common resolvent contour as the mesh is refined; a universal $\sqrt{\sigma}$ law (or any exponent inferred from a finite log-log fit); and a canonical global Hénon operator, zeta factor, von Mangoldt trace, Hilbert–Polya construction, Riemann-zero identification, or proof of RH. All numerical rows are frozen source diagnostics and are not asymptotic evidence. The exact route verdict is Route A = `G0` and Route B = `STOP_SCOPED`.

2 The postcritically finite two-band map

Let

$$p(u) = u^3 - 2u^2 + 2u - 2, \quad u_c = 1.5436890126920764\dots, \quad r = u_c - 1. \quad (1)$$

The map and its invariant interval are

$$f(x) = 1 - u_c x^2, \quad J = [-r, 1]. \quad (2)$$

The symbolic source certificate records

$$f(0) = 1, \quad f(1) = -r, \quad f(-r) = r, \quad f(r) = r. \quad (3)$$

Define the two closed bands (with their common endpoint understood in the usual measure-zero convention)

$$B_0 = [-r, r], \quad B_1 = [r, 1]. \quad (4)$$

Monotonicity on either side of zero and (3) give

$$f(B_0) = B_1, \quad f(B_1) = B_0. \quad (5)$$

The source paper also records the normalized invariant density h and the L^1 sign mode

$$g = h(\mathbf{1}_{B_0} - \mathbf{1}_{B_1}), \quad \mathcal{P}g = -g, \quad (6)$$

in the invariant-density normalization. Equation (6) is an L^1 statement. It does not assert that -1 is isolated on a selected strong space.

3 Exact aligned Ulam inheritance

Definition 3.1 (Exact cell-overlap matrix). Let $\mathcal{D}_h = \{D_i\}_{i=1}^N$ be a finite interval partition of J . The row-stochastic exact cell-overlap matrix is

$$(P_h)_{ij} = \frac{\text{Leb}(D_i \cap f^{-1}(D_j))}{\text{Leb}(D_i)}. \quad (7)$$

We call the partition aligned when r is a cell boundary. Source cells are split at the turning point and inverse-branch breakpoints before the overlap integrals are evaluated.

Theorem 3.2 (Aligned block and sign mode). *For every aligned finite partition, after ordering cells by band,*

$$P_h = \begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}. \quad (8)$$

If $s_i = +1$ on B_0 cells and $s_i = -1$ on B_1 cells, then

$$P_h s = -s. \quad (9)$$

In particular, $-1 \in \text{spec}(P_h)$ in exact arithmetic.

Proof. Every source cell belongs to one band. By (5), its image belongs to the opposite band, so an overlap between cells in the same band is zero. This is exactly the anti-diagonal form (8). Every nonzero row therefore sends its mass to cells carrying the opposite sign; multiplication by P_h changes s to $-s$, which proves (9). No uniform mesh assumption is used. $\square$

Remark 3.3. The theorem is finite-dimensional and normalization-specific. It does not identify the vector s with a right eigenfunction of an arbitrary Lebesgue reference Perron–Frobenius operator, and it does not control the rest of the finite spectrum or its limit as $h \rightarrow 0$.

4 Crossing-cell projection defect

Suppose a single cell D_* crosses r . Define the band fraction and the projected sign by

$$q_i = \frac{\text{Leb}(D_i \cap B_0)}{\text{Leb}(D_i)}, \quad s_i^{(h)} = 2q_i - 1. \quad (10)$$

For an aligned cell, $q_i \in \{0, 1\}$. For the crossing cell, $q_* = \theta \in (0, 1)$. The elementary identity below is the exact local phase law.

Proposition 4.1 (Local phase defect). *For $\theta \in [0, 1]$,*

$$1 - (2\theta - 1)^2 = 4\theta(1 - \theta). \quad (11)$$

If the crossing cell has width h , its width-weighted defect is $4h\theta(1 - \theta)$.

Table 1: Frozen phase-scan counts and ranges. These values are diagnostics.

quantity	value
resolutions	256, 512, 1024, 2048
phases per resolution	33
total rows	136
aligned / crossing rows	4/132
crossing L_h range	1.14×10^{-4} to 7.71×10^{-3}
maximum crossing distance from -1	4.91×10^{-3}
maximum row-sum error	6.4×10^{-9}
maximum stationary L^1 residual	1.9×10^{-13}

Proof. Expand $(2\theta - 1)^2 = 4\theta^2 - 4\theta + 1$ and subtract from one. Multiplying by h gives the second assertion. $\square$

The global diagnostic used by the source is the stationary same-band mass

$$L_h = \sum_i \pi_i q_i \sum_j (P_h)_{ij} q_j + \sum_i \pi_i (1 - q_i) \sum_j (P_h)_{ij} (1 - q_j), \quad (12)$$

where π is a stationary cell-mass vector. If the partition is aligned, the anti-diagonal theorem makes $L_h = 0$ exactly. When a cell crosses r , L_h depends on the stationary weights and all other cells; it is not equal to the local expression in every discretization. This distinction matters: the local identity is a theorem, while the global phase scan is an observed finite quantity.

5 Frozen numerical protocol

The external source uses analytic inverse-branch overlap assembly, row normalization, stationary iteration, and a tracked eigenvalue nearest -1 . The retained deterministic scan has four resolutions

$$N \in \{256, 512, 1024, 2048\}$$

and 33 translated phases at each resolution, together with one snapped aligned phase. Thus there are 136 rows: 4 aligned and 132 crossing. The aligned rows have zero projected same-band mass and an exact sign residual up to floating assembly error. Crossing rows have positive projected mass and a nonzero finite displacement of the tracked near- -1 eigenvalue.

The source also contains direct, split, and cycle-constrained noisy matrices. Their fitted slopes over a finite range are deliberately not promoted here. The boundary convention and kernel change the finite intercepts, and no common strong-space perturbation theorem is supplied.

6 Overlap audit and route decision

RH-3 uses the same band-merging map to prove a continuum parity eigenmode, periodogram limits, and conditional two-step operator decompositions. It does not contain the arbitrary aligned finite-Ulam block theorem or the crossing-cell identity. RH-10 studies exact periodic counts, exponentially close boundary cycles, noncommuting long-cycle/noise limits, and a parity-renormalized determinant; it does not state the phase-local Ulam defect. RH-55 proves midpoint-Ulam strong-weak contour transfer for a conditioned folded-Gaussian kernel; it does not prove deterministic PCF sign inheritance for arbitrary aligned partitions. The overlap is therefore structural context, not duplication.

Route A is GO for Theorems 3.2 and 4.1, together with the source-locked finite protocol. Route B is STOP_SCOPED: the first missing bridge is a common strong-space projector/resolvent

theorem connecting the finite matrices to a continuum operator. In particular, no finite scan proves a universal noise exponent, an isolated continuum resonance, or any arithmetic/RH statement.

7 Conclusion

The exact content of RH-367 is a reusable finite-volume fact: a genuine band-boundary alignment preserves a two-cycle sign mode in every exact cell-overlap matrix, while a crossing cell has an explicit phase defect. The global leakage and near- -1 displacement are useful diagnostics for choosing what must be proved next, but they are not substitutes for a strong-space limit theorem. The physical route coordinate remains `actual_same_clock_unnormalized_head_transport_open`; all five RH gates remain false/open.

References

- [1] Cyclic Ulam Map, source package and proof bundle, 2026. Frozen commit `e7d21f646498d77e1c3213d1e4f35dc8466038ff`.
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- [3] RH-10, *Parity-Renormalized Long-Cycle Determinants*, repository artifact.
- [4] RH-55, *Strong-Weak Adaptive Cutoff Transfer for Intrinsic Riesz Factors*, repository artifact.
- [5] G. Froyland, “On Ulam approximation of isolated spectrum and eigenfunctions of hyperbolic maps,” *DCDS* 17 (2007), 671–689.